

- 4 A student intends to create a web application that allows users to upload image files. To add a layer of security, she wants to encrypt each image filename using one of the two cipher methods described below.

Task 4.1

This variant of the **autokey cipher** applies shifts to characters based on both a keyword and the previously encrypted characters.

Each letter in the keyword corresponds to a shift value: ‘a’ means no shift, ‘b’ means a shift of 1, up to ‘z’ which means a shift of 25 (note that the case of the letters in the keyword does not matter). The keyword is used to encrypt the first few characters of the main text. After the keyword is exhausted, each subsequent letter is encrypted using the previous encrypted letter. Non-alphabetical characters are not encrypted.

For example, if we were to encrypt the text “Hello everybody!” using the keyword “key”, we obtain the result “Rijui mhlcabpsq!”:

Text	H	e	l	l	o		e	v	e	r	y	b	o	d	y	!
Encrypt using	k	e	y	j	u		i	m	h	l	c	a	b	p	s	
Result	R	i	j	u	i		m	h	l	c	a	b	p	s	q	!

If the keyword “keyword” instead, the result would be “Rijhc vyctrsgjh!”.

Text	H	e	l	l	o		e	v	e	r	y	b	o	d	y	!
Encrypt using	k	e	y	w	o		r	d	y	c	t	r	s	g	j	
Result	R	i	j	h	c		v	y	c	t	r	s	g	j	h	!

Write a function `task4_1(main_text, keyword)` that:

- takes two non-empty strings, `main_text` and `keyword`
- applies the cipher described above to `main_text` and returns the resultant string.

Test your function by printing the output for the two examples above.

Save your program code as

TASK4_1_<your name>_<centre number>_<index number>.py [7]

Task 4.2

A **substitution cipher** replaces each letter in the main text with another letter.

The key is a mapping of each letter of the alphabet to another letter, e.g. if the key is “qwertyuiopasdfghjklzxcvbnm”, then “a” in the main text is replaced by “q”, “b” is replaced by “w”, “c” is replaced by “e”, “d” is replaced by “r”, and so on.

[ASRJC/2024/Q4 Cont']

Both upper and lower case letters should be handled, and non-alphabetical characters should be left unchanged.

Using the key “qwertyuiopasdfghjklzxcvbnm” (the letters in the 3 rows of your keyboard), the text “Life is hard. Why not make it harder?” becomes “Soyt ol iqkr. Vin fgz dqat oz iqkrtk?”.

Write a function `task4_2(main_text, key)` that:

- takes a string `main_text` and a 26-letter string `key`
- applies substitution cipher to `main_text` and returns the resultant string.

Test your function by printing the output for the example above, as well as the output of the text “I'll be 100% ready to take the 'A' Levels!”, both using the key “qwertyuiopasdfghjklzxcvbnm”.

Add your program code to

`TASK4_2_<your name>_<centre number>_<index number>.py` [4]

Task 4.3

Write a function `task4_3` that returns a random 26-letter key for the substitution cipher, where all the letters in the key are distinct letters from ‘a’ to ‘z’. You should use basic loops and conditionals to achieve this. Do **not** use the `shuffle()` method from the `random` module.

Add your program code to

`TASK4_3_<your name>_<centre number>_<index number>.py` [3]

Task 4.4

The student eventually decides to use the substitution cipher to encrypt filenames.

Create a single-page web application that allows users to upload image files securely.

- Users should be able to upload image files to an `/uploads` folder in the server.
- Ensure that the filename is safe and free from potentially dangerous characters.
- Filename encryption:
 - Users can enter a 26-letter substitution cipher key in a text box.
 - If no key is provided (or length of key is incorrect), a random key would be created.
 - The filename (excluding the extension) is encrypted using a substitution cipher with the key.
- Upon successful form submission, the web page should indicate that the upload is complete, showing the encrypted filename (with its extension) along with the cipher key used.
- Store the timestamp (current system datetime) and the encrypted filename (with its extension) in a comma-separated text file `uploads.txt`.
- When the page loads, for the case where there are existing entries in the text file, display a table with 3 columns: the first column showing the timestamp, the second showing the encrypted filename (with file extension) and the third showing the corresponding image.

[ASRJC/2024/Q4 Cont']

Test the web application by uploading the 2 image files provided (with extensions .jpg, .png).
Use the cipher key “qwertyuiopasdfghjklzxcvbnm” for all uploads.

Save the project files (including “uploads.txt”) and sub-folders for this task in a folder named

TASK4_4_<your name>_<centre number>_<index number>

[10]